Pharma Replenishment Agent Rule Book

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How to read this rule book

Each role's agent acts on that role's behalf, within the rules below. Humans set the rules once and handle only the exceptions. All thresholds are demo defaults; each role owner tunes their own.

Every rule has four parts:

1. **When** – the trigger that wakes the agent (a schedule or an event).
2. **Condition** – what the agent checks.
3. **Agent action** – what it does when the condition holds.
4. **Escalate when** – the cases it hands to a human, with its reasoning attached.

Each rule also carries an autonomy level, so you can show trust growing over time:

Level	Name	What the agent does	What the human does
L1	Assisted	Recommends and explains	Clicks approve on every item
L2	Supervised	Acts within thresholds	Approves exceptions only
L3	Autonomous	Acts end to end	Reviews the dashboard and audit log

Pharmacy agent

The Pharmacy agent replaces the pharmacist logging in to order. It watches billing and stock, forecasts demand and raises orders to the dealer.

Rule	When	Condition	Agent action	Escalate when	Level
PH-01 Reorder	Nightly at 23:00, and on every bill	Stock on hand < reorder point (7 days of forecast demand + safety stock)	Raise a purchase order to the dealer for 14 days of cover	Never for routine items	L3

Rule	When	Condition	Agent action	Escalate when	Level
PH-02 Forecast	Every Sunday	8+ weeks of sales history exists	Refresh the 4-week demand forecast per medicine	Forecast error > 30% for 2 weeks in a row	L2
PH-03 Spike	On every bill	Daily sales > 3× the 4-week average	Hold the auto-order and flag the spike	Always – the pharmacist confirms seasonal or one-off	L1
PH-04 Expiry	Daily	Batch expires within 90 days	Stop reordering it and suggest return to dealer	Stock value at risk > ₹10,000	L2
PH-05 Scheduled drugs	On order creation	Item is Schedule H1 or X	Draft the order only	Always – the licensed pharmacist approves	L1
PH-06 Order cap	On order creation	Order value > ₹50,000	Draft the order only	Always	L1

Dealer agent

The Dealer agent fulfils pharmacy orders from its own stock and rolls up any shortfall into one order to the C&F.

Rule	When	Condition	Agent action	Escalate when	Level
DL-01 Fulfil	On each pharmacy order	Dealer stock covers the full quantity	Confirm the order and schedule dispatch	Never	L3
DL-02 Pharmacy credit	On each pharmacy order	Pharmacy outstanding + order ≤ credit limit, no dues > 45 days	Accept the order	Limit exceeded or overdue dues – dealer decides	L2
DL-03 Partial fill	On each pharmacy order	Stock covers only part of the quantity	Ship what is available, back-order the rest	Fill rate < 50% on a critical medicine	L2
DL-04 Roll-up	Daily at 18:00	Shortfalls + own reorder needs exist	Consolidate into one order to the C&F	Order value > ₹5 lakh	L2
DL-05 Allocation	When stock is short across pharmacies	Demand > available stock	Allocate by forecast share, critical drugs first	Any pharmacy gets < 50% of its order	L1

C&F agent

The C&F agent auto-approves dealer orders that pass all five checks below; the C&F person sees only the orders that fail one.

Rule	When	Condition	Agent action	Escalate when	Level
CF-01 Auto-approve	On each dealer order	CF-02 to CF-05 all pass	Approve and create the dispatch note	Any check fails	L2
CF-02 Dealer credit	On each dealer order	Dealer outstanding + order ⩽ credit limit	Pass	Limit exceeded – C&F decides or asks for advance payment	L2
CF-03 Stock	On each dealer order	Warehouse stock covers the order	Pass; reserve the stock	Shortage – split the order or raise to manufacturer	L2
CF-04 Forecast fit	On each dealer order	Quantity within ±20% of the dealer's forecast	Pass	Outside the band – possible hoarding or a real spike	L1
CF-05 Expiry (FEFO)	At picking	Every batch shipped has ⩾ 6 months of shelf life	Pick first-expiry-first-out	No batch meets shelf life	L2
CF-06 Replenish	Daily at 20:00	Warehouse stock < 15 days of total dealer demand	Raise a stock transfer request to the manufacturer	Request > monthly quota	L2

Manufacturer agent

The Manufacturer agent confirms C&F stock requests within each region's monthly quota and feeds aggregated demand into production planning.

Rule	When	Condition	Agent action	Escalate when	Level
MF-01 Quota approve	On each C&F request	Cumulative month's dispatch ⩽ the C&F's monthly quota	Approve and schedule the plant dispatch	Quota exceeded	L2
MF-02 Finished goods	On each C&F request	Finished-goods stock covers the request	Allocate stock	Shortage across regions – supply planner allocates	L1

Rule	When	Condition	Agent action	Escalate when	Level
MF-03 Demand signal	Every Monday	Forecasts from all C&Fs received	Publish the 8-week demand plan to production	Total demand > plant capacity by > 10%	L2
MF-04 Batch release	Before dispatch	Batch has QA release	Dispatch	Never ship – QA hold always goes to QA head	L1
MF-05 Quota review	Monthly, 1st day	Actual vs quota per C&F	Propose next month's quotas	Any C&F change > 15% – sales head approves	L1

Main agent (orchestrator)

The main agent owns the order's end-to-end state: it routes events between agents, enforces timeouts and keeps the audit trail.

Rule	When	Condition	Agent action	Escalate when
OR-01 Routing	On every order event	Event has a valid next owner	Hand the order to the next agent	Unknown state or missing owner
OR-02 Retry	On a failed agent or API call	Fewer than 3 attempts	Retry with 2, 4, 8 minute back-off	3rd failure – IT support
OR-03 Approval timeout	Every 30 minutes	Human escalation unanswered for 4 hours	Remind; after 8 hours, send to the backup approver	Critical medicine waiting > 8 hours
OR-04 Audit	On every decision	Always	Log who (agent or human), rule ID, inputs and reason	Never
OR-05 Kill switch	Anytime	A role owner pauses their agent	Drop that role back to L1	Never

Rule	When	Condition	Agent action	Escalate when
OR-06 Daily summary	Daily at 08:00	Always	Email each role: auto-approved count, escalations, pending	Never

Escalation matrix

Escalations arrive as a message with Approve and Reject buttons and the agent's reasoning; no login is needed.

Raised by	Goes to	Typical triggers	Channel	Response SLA	Backup after 8 h
Pharmacy agent	Pharmacist	Spike, Schedule H1/X, order > ₹50,000	WhatsApp / app push	4 hours	Pharmacy owner
Dealer agent	Dealer owner	Credit breach, low fill rate, roll-up > ₹5 lakh	WhatsApp / email	4 hours	Dealer accounts head
C&F agent	C&F manager	Credit, stock shortage, forecast band breach	Email / Teams	4 hours	Regional sales manager
Manufacturer agent	Supply planner	Quota exceeded, regional shortage, capacity gap	Email / Teams	8 hours	Head of supply chain
Manufacturer agent	QA head	Batch on QA hold	Email	24 hours	Plant head
Main agent	IT support	3 failed retries, unknown order state	Email / ticket	1 hour	IT lead

Demo storyline

Three scenarios show the full story: routine orders run without anyone, exceptions reach the right person, and autonomy grows with trust.

1. **Routine order – no human touches it.** Paracetamol drops below the reorder point at 23:00. PH-01 raises the order, DL-01 fulfils part, DL-04 rolls up the rest, CF-01 auto-approves, MF-01 confirms. Show the audit log with every rule ID.

2. **Exception – one human, one tap.** A dealer orders 3× its forecast of an antibiotic. CF-04 fails, so the C&F manager gets a message with the agent's reasoning and taps Reject. Everything else in the chain kept moving.
3. **Trust grows – level change.** Show CF-04 at L1 on day 1. After the dashboard shows 4 weeks of agent recommendations matching human decisions, the C&F owner moves it to L2. Then pause the C&F agent with OR-05 to show humans stay in control.

Close on OR-06: the daily summary email that replaces four manual logins.